Self-Supervised Representation Learning for Organoid Clustering

Shivam Gupta[1]

TU Dresden, Germany

Abstract

In current biological and medical research, working with microscopic images is routine work for scientists. It enables large scale collection of organoid images that are morphologically heterogeneous in nature. An important challenge is therefore to identify groups of organoids that share similar visual and structural characteristics. In this work, I investigated self-supervised representation learning for organoid image analysis using Masked Siamese Networks (MSN) [1]. The proposed approach learns image embeddings from unlabeled data by maintaining consistency between masked and unmasked views of the same image. Subsequently, these representations were used to assess the similarity between organoid images and to support correlation-based clustering. The learned embeddings were evaluated quantitatively and qualitatively, demonstrating that MSN produces meaningful representations for clustering organoids.

1 Introduction

Organoids are widely used in biological and medical research to study development, disease mechanisms, and cellular organization. Advances in microscopy now allow scientists to acquire large collections of organoid images in a single experiment, revealing substantial morphological variability between samples (cf. Figure 1). An important challenge is therefore to identify groups of organoids that exhibit similar visual and structural characteristics, as manual inspection does not scale to such data volumes. Due to the high heterogeneity of organoid morphologies and the lack of clearly defined classes, this problem is addressed more naturally as a clustering task than supervised classification.

In this project, I adopt Masked Siamese Networks (MSN), a self-supervised representation learning framework that combines siamese learning with masked-view consistency. MSN operates on two augmented views of the same image: one view is partially masked by randomly dropping image patches, while the other remains fully unmasked. Both views are processed by a shared Vision Transformer (ViT) encoder to produce image embeddings. MSN architectures do not directly predict pixel values (or tokens) for the masked patches. Instead, the loss is applied directly to the output corresponding to the [CLS] token of the encoder [3]. The training objective is simple: ensure that the embedding from the masked view matches the embedding from the unmasked view. This forces the model to denoise at the representation level rather than at the pixel level.

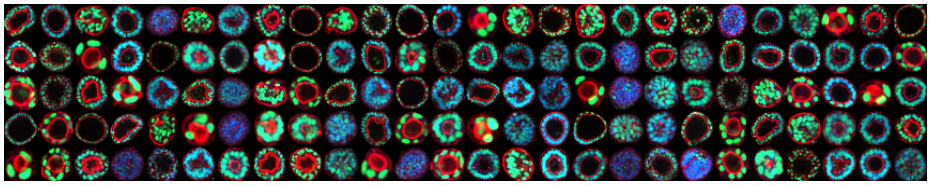


Fig. 1: Shown are 130 organoid images of pancreatic progenitor organoids derived from human pluripotent stem cells, scaled to a common size for visualization. Adapted from [2].

This design encourages the model to learn high-level, semantically meaningful representations that focus on what the object is rather than low-level details like texture or lighting. The learned representations are robust to missing visual information, making them particularly suitable for downstream tasks like similarity estimation and clustering.

The focus is on using MSN to learn representations of organoid images that allow the calculation of similarity in appearance and morphology. These learned embeddings serve as the basis for defining correlation costs between pairs of organoid images. I use correlation clustering approach that aggregates potentially conflicting pairwise relations without requiring a predefined number of clusters. Unlike methods such as k-means that need the number of clusters specified in advance, correlation clustering automatically infers cluster structure from pairwise similarities, making it well-suited for organoid analysis where the number of distinct morphological groups may not be known beforehand. The quality of the learned representations on labeled benchmark datasets is evaluated, complemented by qualitative analysis on unlabeled organoid image collections.

2 Related Work

Self-supervised learning (SSL) has emerged as a powerful approach for learning visual representations from unlabeled data. Early contrastive methods like SimCLR [4] work by enforcing invariance across multiple augmented views of the same image using negative samples. More recent joint-embedding architectures, including SwAV [5] and DINO [6], take a different approach by avoiding explicit negatives and relying instead on clustering-based objectives or momentum encoders to prevent representation collapse. These methods have proven effective at learning semantically meaningful embeddings that generalize well to downstream tasks. MSN [3] build on this class of joint-embedding methods by adding masked-view consistency, without needing unmasked anchor views. Results on ImageNet low-shot classification show that MSN achieves strong performance with far fewer labeled samples, outperforming larger state-of-the-art models while using substantially fewer parameters (Figure 2). This robustness to limited supervision makes MSN especially attractive for domains like biomedical imaging, where labeled data are hard to come by.

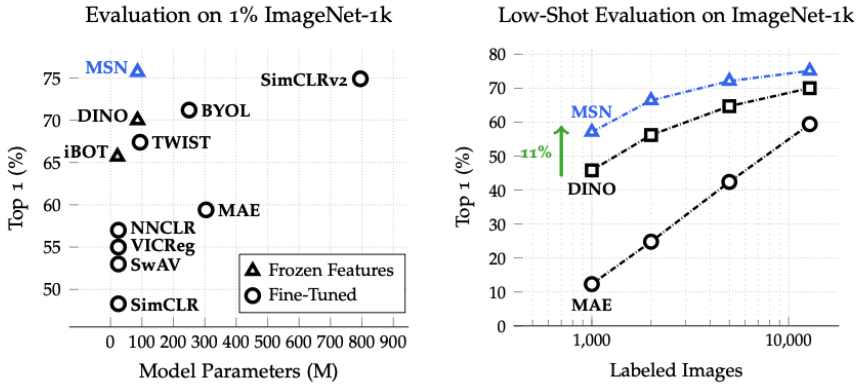


Fig. 2: **Low-shot evaluation of self-supervised models pre-trained on ImageNet-1K. Left:** Performance on the 1% ImageNet classification task, where MSN achieves good accuracy while using significantly fewer parameters. **Right:** Low-shot classification accuracy as a function of labeled samples, showing that MSN achieves competitive performance with substantially fewer labels compared to other masked approaches. Adapted from [3].

A different line of work explores masked image modeling, where parts of the input image are removed and the model learns to reconstruct what's missing. Recent techniques like Masked Autoencoders (MAE) [7] apply this idea to vision transformers, similar to MSN. However, these methods typically use pixel or token level reconstruction losses, which often focus on low-level visual details and may need substantial fine-tuning for semantic tasks. MSN takes a different approach by avoiding explicit reconstruction altogether. Instead, it performs denoising implicitly at the representation level by matching embeddings of masked and unmasked views. This leads to better computational efficiency and stronger semantic representations.

In organoid image analysis, earlier work has tackled clustering by explicitly modeling pairwise similarity between images and consolidating these relationships through correlation clustering. Presberger et al. [2] frame organoid clustering as a correlation clustering problem, where correlation costs come from either a partial quadratic assignment problem (PQAP) or from supervised twin networks trained on annotated image pairs. While these approaches produce accurate clusters, they need labeled data to learn similarity models. My work takes a different path by using a self-supervised approach, where similarity is implicitly encoded in learned embeddings from MSN, and labels are only used for evaluation. I then use these embeddings to define correlation costs for clustering, making it possible to analyze large collections of organoid images without needing manually annotated image pairs. As shown in Figure 2, MSN demonstrates strong performance in low-shot learning scenarios. On the 1% ImageNet classification task, MSN achieves competitive accuracy while using substantially fewer

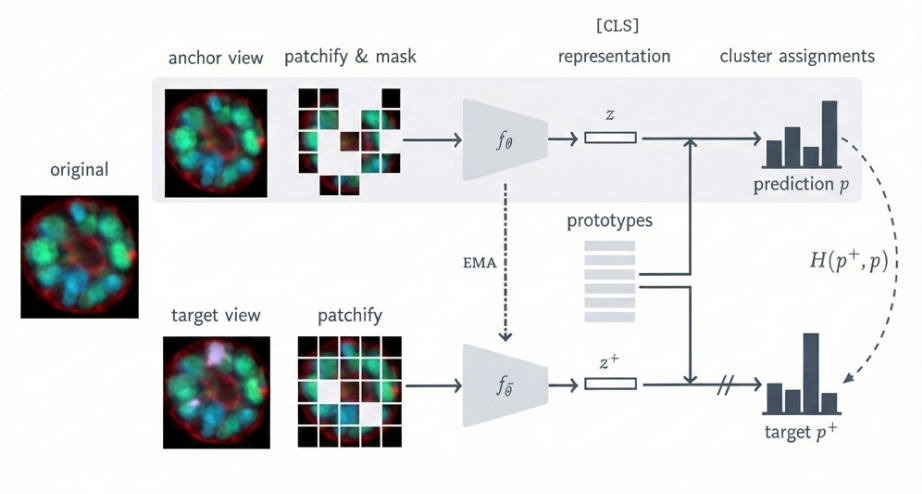


Fig. 3: Overview of the Masked Siamese Networks (MSN) training framework. A masked anchor view and an unmasked target view are encoded using Vision Transformers. Learning is driven by aligning prototype assignments of the two views. Adapted from [3]

parameters than prior methods. Moreover, MSN requires significantly fewer labeled samples to reach comparable performance with masked auto-encoder approaches, making it particularly suitable for domains where labeled data are limited.

3 Prerequisites

Problem Formulation. Let $D = \{x_i\}_{i=1}^{U}$ denote a large collection of unlabeled organoid images and $S = \{(x_i^s, y_i)\}_{i=1}^{L}$ a much smaller labeled dataset, with $L \ll U$. The labeled dataset may partially overlap with the unlabeled collection. The objective of this project is to learn image representations from the unlabeled data D that capture meaningful morphological similarities between organoids. These representations are then used to define pairwise similarities and to support clustering, while the labeled dataset S can be used for evaluation.

Dataset. The datasets used in this work consist of microscopy images of single organoids with varying morphologies. The unlabeled dataset contains a large number of organoid images and is used for self-supervised representation learning. In addition, labeled benchmark datasets are provided, where organoid images are grouped into morphological classes based on expert annotation. These labeled datasets can be used exclusively to evaluate clustering performance.

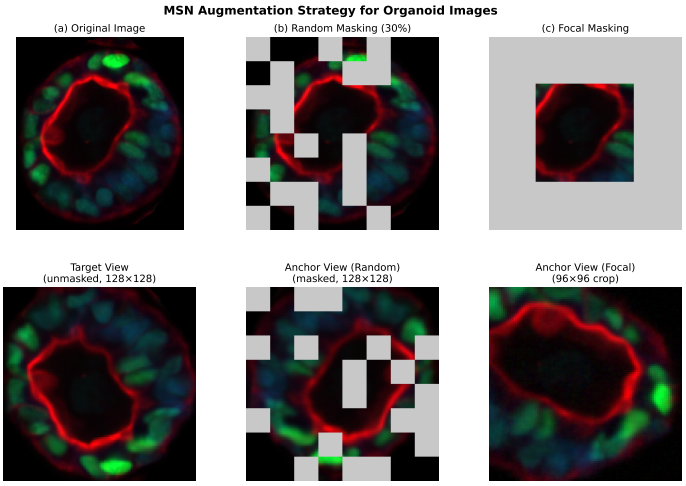


Fig. 4: Data augmentation and masking strategies for organoid MSN training. (a) Original organoid image. (b) Random masking removes 30% of patches across the image. (c) Focal masking crops to a 50–85% scale region. Bottom row shows the resulting views after applying augmentations: the unmasked target view (128×128), anchor view with random masking (128×128), and anchor view with focal crop (96×96). Augmentations include random flips, rotations (90°), elastic deformations, Gaussian noise, and Gaussian blur.

4 Masked Siamese Networks (MSN) Model

MSN is a self-supervised representation learning method that learns image embeddings by enforcing consistency between masked and unmasked views of the same image (Figure 3).

Siamese Setup and Input Views. For each input image, random data augmentations generate two types of views: an unmasked *target view* and M *anchor views*. The target view remains fully visible, while anchor views are partially masked by randomly dropping image patches. This encourages the model to learn robust, high-level representations invariant to missing information.

Vision Transformer Encoder. Both views are processed by Vision Transformer (ViT) encoders. Each image is split into non-overlapping patches, linearly embedded, and augmented with positional encodings. A `[CLS]` token aggregates global information, and its output is the final embedding. The target encoder parameters are updated via exponential moving average of the anchor encoder, stabilizing training and preventing representation collapse.

Prototype-Based Predictions. MSN uses K learnable prototype vectors $q \in \mathbb{R}^{K \times d}$. For each anchor embedding $z_{i,m}$, a probability distribution over prototypes is computed via softmax:

$$p_{i,m} = \text{softmax}\left(\frac{z_{i,m} \cdot q}{\tau}\right),$$

where τ controls distribution sharpness. Target predictions p_i^+ use a sharper temperature.

Training Objective and ME-MAX Regularization. MSN processes batches of B images with M masked views each, minimizing cross-entropy loss:

$$\mathcal{L}_{\text{align}} = \frac{1}{MB} \sum_{i=1}^{B} \sum_{m=1}^{M} H\big(p_i^+, p_{i,m}\big)\,,$$

where $p_{i,m}$ and p_i^+ are prototype distributions for masked and unmasked views respectively. To prevent collapse to few prototypes, MSN adds mean-entropy maximization (ME-MAX):

$$\mathcal{L} = \mathcal{L}_{\text{align}} - \lambda H(\bar{p}),$$

where

$$\bar{p} = \frac{1}{MB} \sum_{i=1}^{B} \sum_{m=1}^{M} p_{i,m}$$

is the average prototype assignment and $\lambda > 0$ controls regularization strength. This encourages uniform prototype usage. After training, the prototype head is discarded. The ViT encoder extracts embeddings used to compute pairwise similarities for clustering.

5 Methodology

5.1 Training MSN on Organoid Images

I trained the MSN encoder on 80,000 unlabeled organoid images. A critical adaptation was the choice of data augmentations. After analyzing the original MSN augmentation strategy, I found that color distortion (standard for natural images) was unsuitable for organoid microscopy, where color is a primary distinguishing feature between morphological classes. I therefore excluded color jitter and instead used random cropping, horizontal and vertical flips, random rotations, Gaussian noise, and elastic deformations.

Through iterative experimentation, I observed that the organoid dataset's substantial variability in image sizes caused smaller organoids to become unrecognizable after aggressive augmentations. To address this, I reduced the input resolution from 224×224 to 128×128 pixels and adjusted augmentation parameters. I validated configurations on dataset subsets for faster iteration. The final setup used 128×128 resolution, moderate augmentation aggressiveness, and focal cropping with scale range 0.5–0.85, and introducing patch drop in range 0.2 to 0.4, and trained for 100 epochs with batch size 64. This configuration balanced augmentation-invariant learning while preserving distinctive morphological features.

5.2 Feature Extraction

After training, I used the frozen MSN encoder to extract features from Test-100 and Test-30. Each image was preprocessed by resizing to 146×146 pixels, center cropping to 128×128 pixels, and normalizing with the statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]). The [CLS] token output from the Vision Transformer was extracted as the image embedding. All embeddings were L2-normalized to ensure cosine similarities lie in $[-1, 1]$, where values near 1 indicate high similarity.

5.3 Correlation Clustering

I formulated clustering as a correlation clustering problem. For each image pair (i, j), I computed correlation costs from normalized embeddings:

$$c_{ij} = m - \cos(g(x_i), g(x_j)),$$

where $g(x_i)$ is the MSN embedding and m is a margin hyperparameter. After experimentation, I set $m = 0.20$, meaning pairs with cosine similarity above 0.20 receive negative costs (should cluster together), while dissimilar pairs receive positive costs (should separate).

I solved this using a greedy union-find algorithm: process pairs by increasing cost, merge pairs with negative cost. This automatically determines the number of clusters from pairwise similarities, unlike k-means which requires specifying cluster count beforehand, particularly valuable for organoid analysis where the number of morphological groups is unknown.

6 Results

6.1 Dataset and Evaluation Setup

I conduct experiments following the dataset split from Presberger et al. The evaluation uses Test-100 (100 images, 10 classes, 10 images per class) for in-distribution evaluation and Test-30 (30 images, 3 different classes, 10 images per class) for out-of-distribution generalization. All datasets include ground-truth cluster labels provided by domain experts. Features are extracted using the trained MSN encoder and evaluated via correlation clustering using Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and clustering accuracy.

6.2 Clustering Performance

On Test-100, the inferred clustering closely matches ground-truth partitioning with ARI of 0.6705, NMI of 0.8065, and accuracy of 0.790. The method correctly recovers 10 clusters, indicating that learned representations preserve class-level structure for organoid morphologies seen during training.

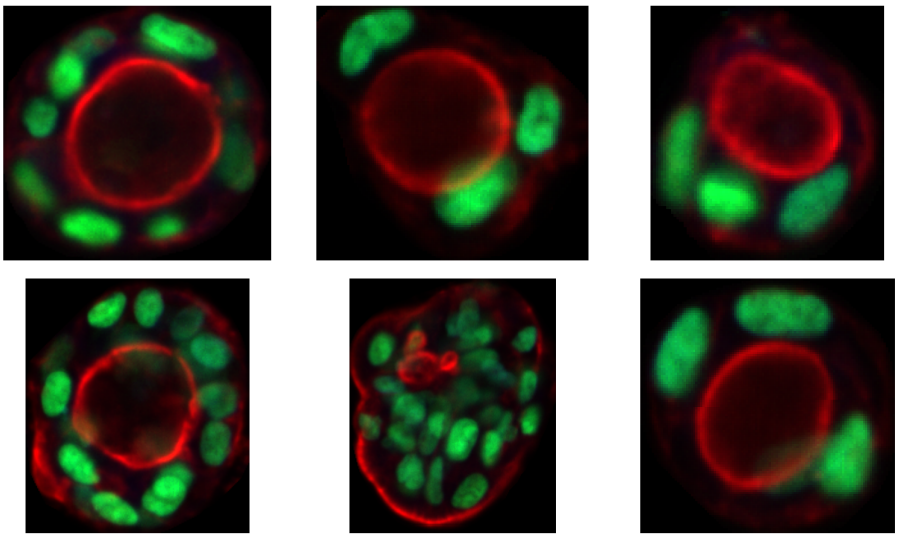


Fig. 5: Representative organoid images from a single cluster obtained using MSN embeddings. Five out of six images belong to the same annotated class.

On Test-30 with previously unseen organoid classes, performance decreases but remains robust: ARI of 0.5123, NMI of 0.6083, and accuracy of 0.7667, again recovering the correct number of clusters (3). This drop reflects the increased difficulty of generalizing to novel morphologies, while demonstrating that representations capture transferable structural information.

Overall, MSN-based representations support meaningful clustering of organoid images for both seen and unseen classes without requiring the number of clusters as input.

6.3 Qualitative Analysis

Figure 5 shows representative examples from a single cluster obtained using MSN embeddings. Five out of six displayed organoids belong to the same morphological class, indicating that learned embeddings capture biologically meaningful similarity. The remaining image differs mildly in appearance, suggesting the cluster tolerates intra-class variability commonly observed in organoid morphology. This visualization supports the quantitative results and demonstrates MSN's ability to group organoids with similar structural characteristics.

7 Conclusion

In this work, I investigated Masked Siamese Networks (MSN) for self-supervised representation learning in organoid image clustering. The objective was to learn

meaningful embeddings from unlabeled organoid images that capture morphological similarities without manual annotations during training.

The results demonstrate that MSN successfully learns representations enabling effective clustering. On Test-100, I achieved strong performance (ARI: 0.6705, NMI: 0.8065, accuracy: 0.7900), and the approach generalizes reasonably to unseen morphologies on Test-30 (ARI: 0.5123, NMI: 0.6083, accuracy: 0.7667). Both datasets recovered the correct number of clusters automatically.

The MSN-based approach offers key advantages: it eliminates the need for labeled training data, learns high-level semantic representations robust to missing visual information, and integrates seamlessly with correlation clustering to produce biologically meaningful groupings. The qualitative analysis confirms that clusters predominantly contain organoids from the same morphological class while tolerating natural intra-class variability.

However, the performance indicates room for improvement in generalization to novel morphologies. Based on my research and experiments, I identified that excluding color augmentation—while necessary to preserve biologically meaningful color information, may limit the model's ability to learn more robust invariance. Future work could address this by increasing batch size for more stable training, using more prototypes to capture finer-grained morphological variations, and training for additional epochs to allow better convergence. Semi-supervised approaches leveraging small amounts of labeled data could also be explored.

This project demonstrates that MSN provides an effective framework for self-supervised organoid image clustering. The learned embeddings capture morphological similarity and enable accurate clustering for both seen and unseen classes, offering a scalable solution for analyzing large organoid image collections without extensive manual annotation.

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